

Homework 8 – Reconstructing a musical genome

This assignment can be done individually or in pairs. Please turn in only one copy of your results. Do not forget to list both participants if working as a pair.

The attached file, hw8_kmers.xlsx, contains 5mers from a musical text. The list has two columns, the first with the word, and the second with the number of times that 5mer appeared in the text. This list of words was constructed as follows:

1. All punctuation and spaces were removed from the text.
2. 30 letter reads were randomly sampled from the text. My goal was coverage=4. I obtained 96 reads with 716 letters.
3. From the 96 reads, I made a list of all 6mers and alphabetized the list for convenience in looking up words. I found a total of 3882 kmers, which fall into 425 unique types. My average coverage is thus $3882/425 = 9.13$.
4. Construct the De Bruijn graph from these reads showing the condensed (unbranched) nodes and the connections between nodes. You do not have to draw a graph, a text version is OK (see the example for one approach; anything that is clear is fine). The condensed nodes in this graph should show all the unbranched k-1 overlaps.

Attached to Brightspace assignment

5. Calculate the number of contigs, length of the assembly, and the N50 for your assembly. The length of the assembly is the sum of the lengths of your contigs, not counting duplicate overlapping regions at the ends.

Number of contigs: 70 – once I created the graph, there were many overlaps, and each remaining piece is a contig. I found 70 pieces that are contigs.

Length of assembly: 775 letters – the contig lengths all add up to be 775.

N50: $775/2 = 387.5$; sorted the contigs from longest to shortest and added the lengths and it reached half (387.5) of the total assembly length. The **N50 is 11** because the contigs are short and I reached 50% of the total length once I reached contigs that were 11 letters long. 50% of the assembly is in contigs that are 11 letters or longer.

6. Although the graph above contains all the unbranched segments, you may be able to extend the contigs using external knowledge, in this case, your knowledge of English grammar.

Questions

1. What is the song?

The song is called Rag'n'Bone Man by Human.

2. What are the number of contigs, length of assembly, and N50 for your assembly?

There are 70 contigs, the length of assembly is 775 letters, and the N50 is 11.

3. How does your assembly length compare to the length you would predict from the kmers?

The actual assembly length is 775 letters. The longest contig is 45 letters, and the shortest contig is 6 letters. The predicted length is the unique number of kmers $\times (k-1)$ and $k = 6$. There are 425 rows which means there are 425 unique kmers. $425 \times 5 = 2125$ because $6-1 = 5$. The actual assembly length is 775 letters, so the difference was 1350 letters.

4. Why are the kmer estimates of the length and the sum of the lengths of your contigs?

The K-mer length estimate wants a single sequence where there are unique kmers and each kmer is only used once and there are no overlaps or repeats. The contigs that I have from the graph show repeats, gaps, and branches. When there are repeats and branches, it causes the sequences to split into more pieces, and then you must count those pieces which then makes the sum of the lengths even bigger than the estimate.